Step 1: Install Required Libraries

```
!pip install transformers datasets torch scikit-learn
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.3)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.5
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.2
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->t
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformer
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspe
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hugg
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datase
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0
```

Step 2: Import Libraries

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer
from transformers import AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments
import numpy as np
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

Step 3: Load Dataset

```
dataset = load_dataset("imdb")
print(dataset)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
README.md:       7.81k/? [00:00<00:00, 452kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
plain_text/train-00000-of-00001.parquet: 100%  21.0M/21.0M [00:01<00:00, 18.0MB/s]
plain_text/test-00000-of-00001.parquet: 100%  20.5M/20.5M [00:00<00:00, 36.7MB/s]
plain_text/unsupervised-00000-of-00001.p(…): 100%  42.0M/42.0M [00:00<00:00, 161MB/s]
Generating train split: 100%  25000/25000 [00:00<00:00, 84109.40 examples/s]
Generating test split: 100%  25000/25000 [00:00<00:00, 90743.19 examples/s]
Generating unsupervised split: 100%  50000/50000 [00:00<00:00, 107248.17 examples/s]
DatasetDict({
    train: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['text', 'label'],
        num_rows: 50000
    })
})
```

Step 4: Load Tokenizer

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
config.json: 100%  483/483 [00:00<00:00, 5.91kB/s]
tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 694B/s]
vocab.txt:       232k/? [00:00<00:00, 3.04MB/s]
tokenizer.json:       466k/? [00:00<00:00, 6.12MB/s]
```

```
Step 5: Tokenize the Text
```

```
def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

tokenized_dataset = dataset.map(tokenize_function, batched=True)
```

```
Map: 100%  25000/25000 [00:34<00:00, 773.94 examples/s]
Map: 100%  25000/25000 [00:35<00:00, 624.67 examples/s]
Map: 100%  50000/50000 [01:11<00:00, 694.82 examples/s]
```

Step 6: Set Format for PyTorch

```
tokenized_dataset.set_format("torch",
                             columns=["input_ids","attention_mask","label"])
```

Step 7: Train-Test Split

```
train_dataset = tokenized_dataset["train"]
test_dataset = tokenized_dataset["test"]
```

Step 8: Load Pretrained Transformer Model

```
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

```
model.safetensors: 100%                                                    268M/268M [00:05<00:00, 183MB/s]

Loading weights: 100%                                  100/100 [00:00<00:00, 285.03it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.bias   | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.bias         | MISSING    |
classifier.weight       | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

Step 9: Define Evaluation Metrics

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = np.argmax(pred.predictions, axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

Step 10: Define Training Arguments

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=2,
    eval_strategy="epoch"
)
```

Step 11: Create Trainer

```
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)
```

Step 12: Train the Model

```
import torch
from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification, Trainer, TrainingArguments
import numpy as np
```

```
from sklearn.metrics import accuracy_score, precision_recall_fscore_support

# Step 3: Load Dataset (from S18JIAAm5uv6)
dataset = load_dataset("imdb")

# Step 4: Load Tokenizer (from i_fRZIOq7TgU)
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")

# Step 5: Tokenize the Text (from Fh9lWU4w7ilX)
def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)
tokenized_dataset = dataset.map(tokenize_function, batched=True)

# Step 6: Set Format for PyTorch (from ZKTXGqwa8yJ1)
tokenized_dataset.set_format("torch", columns=["input_ids","attention_mask","label"])

# Step 7: Train-Test Split (from O2mBAw329DAu)
train_dataset = tokenized_dataset["train"]
test_dataset = tokenized_dataset["test"]

# Step 8: Load Pretrained Transformer Model (from f64iEkUl9L2n) - already copied
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)

# Step 9: Define Evaluation Metrics (from meFuqaUL_1sR)
def compute_metrics(pred):
    labels = pred.label_ids
    preds = np.argmax(pred.predictions, axis=1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    acc = accuracy_score(labels, preds)

    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }

# Step 10: Define Training Arguments (from VXRiDm2dANHw) - already copied
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=2,
    eval_strategy="epoch"
)

# Step 11: Create Trainer and Step 12: Train the Model
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)
trainer.train()
```

```
README.md:          7.81k/?  [00:00<00:00,  96.6kB/s]

plain_text/train-00000-of-00001.parquet:  100%                                                    21.0M/21.0M  [00:01<00:00,  14.4MB/s]

plain_text/test-00000-of-00001.parquet:  100%                                                    20.5M/20.5M  [00:00<00:00,  161MB/s]

plain_text/unsupervised-00000-of-00001.p(…):  100%                                                    42.0M/42.0M  [00:01<00:00,  65.5MB/s]

Generating train split:  100%                                                    25000/25000  [00:01<00:00,  29434.92  examples/s]

Generating test split:  100%                                                    25000/25000  [00:00<00:00,  40631.31  examples/s]

Generating unsupervised split:  100%                                                    50000/50000  [00:01<00:00,  50001.43  examples/s]

tokenizer_config.json:  100%                                                    48.0/48.0  [00:00<00:00,  1.91kB/s]

vocab.txt:          232k/?  [00:00<00:00,  1.62MB/s]

tokenizer.json:          466k/?  [00:00<00:00,  1.32MB/s]

Map:  100%                                                    25000/25000  [00:43<00:00,  801.49  examples/s]

Map:  100%                                                    25000/25000  [00:29<00:00,  905.74  examples/s]

Map:  100%                                                    50000/50000  [01:07<00:00,  582.88  examples/s]

Loading weights:  100%                                                    100/100  [00:00<00:00,  450.24it/s,  Materializing  param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_projector.bias     | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
vocab_layer_norm.weight  | UNEXPECTED |
classifier.bias          | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.weight        | MISSING    |
pre_classifier.bias      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[185/3126 2:25:49 < 39:03:35, 0.02 it/s, Epoch 0.12/2]

Epoch	Training Loss	Validation Loss